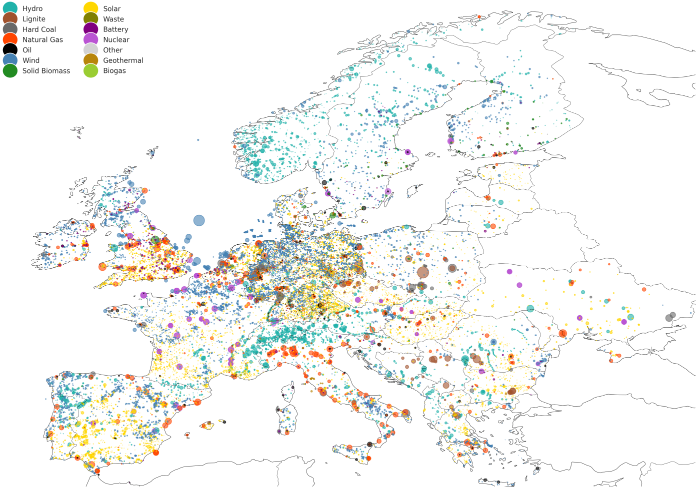
A first modelling quest



Many **power plants** in Europe.

Many **consumers** in Europe.

Interconnected through **power grid**.

How to decide which of these power plants produce at a given time?

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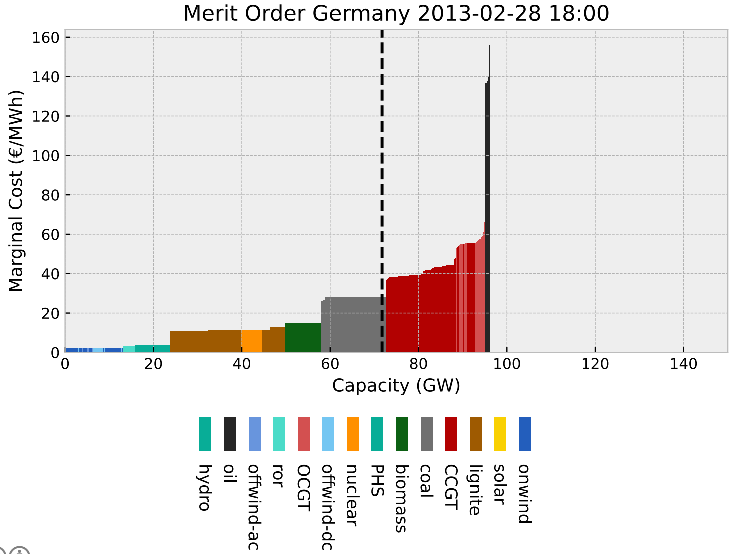
For one market region and one hour we can start with merit order



Sort generators by marginal cost and how much they could produce in a given hour.

Consider wind, solar, hydro availability & outages.

Here:
Assume fixed demand.

Price set by the generator intersecting with demand.

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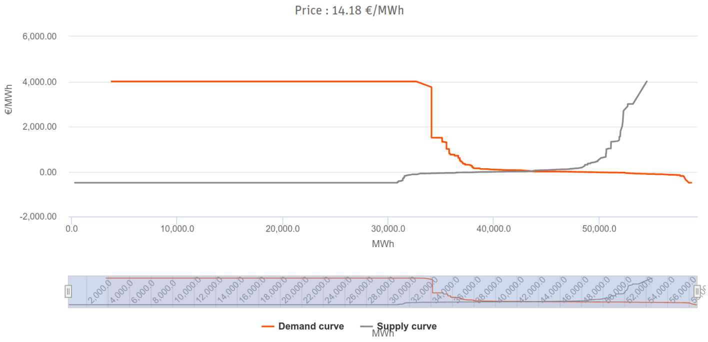
And in reality?

Auction > Day-Ahead > SDAC > DE-LU > 16 October 2024

Last update: 15 October 2024 (12:57:56 CET/CEST)



Displayed aggregated curves include all orders submitted to the operating coupled NEMOs for resp. SDAC or SIDC auction in the concerned bidding zone ; the displayed "Price" information reflects the EPEX SPOT market clearing price of the respective EPEX SPOT auction. For delivery date 26/06/2024, the displayed aggregated curves for SDAC auction in AT, BE, DE, FR and NL include all orders submitted to EPEX SPOT only.

EPEXSPOT day-ahead electricity market

Shown for 1 hour in German market zone.

Demand side has varying willingness to pay, just like supply side (power plants) has varying marginal cost.

Source: https://www.epexspot.com/en/market-results

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All this can be formulated as an optimisation problem!

For a single region and hour, we have s generators with marginal costs o_s (€/MWh) and (available) capacities G_s (MW). We seek to minimise the operational costs of generator dispatch g_s to fulfil the electricity load of d MW. The **optimisation problem**:

$$\min_{g_s} \sum_s o_s g_s \tag{1}$$

such that

$$g_s \leq G_s \tag{2}$$

$$g_s \geq 0 \tag{3}$$

$$\sum_s g_s = d \tag{4}$$

(1) is the **objective function**, (2-4) are inequality and equality **constraints**. g_s are the **optimisation variables** or **decision variables**. G_s, o_s and d are **input parameters**.

Adding a spatial component $i \to$ e.g. multiple market zones

$$\min_{g_{i,s}, f_\ell} \sum_s o_{i,s} g_{i,s}$$

such that

$$g_{i,s} \leq G_{i,s}$$

$$g_{i,s} \geq 0$$

$$\sum_s g_{i,s} - \sum_\ell K_{i\ell} f_\ell = d_i \qquad \text{Kirchhoff Current Law}$$

$$|f_\ell| \leq F_\ell \qquad \text{line limits}$$

$$\sum_\ell C_{\ell c} x_\ell f_\ell = 0 \qquad \text{Kirchhoff Voltage Law}$$

where f_ℓ is the power flow (e.g. into/from other market zones), $K_{i\ell}$ is the incidence matrix and F_ℓ is the transmission line capacity, x_ℓ is the line reactance, and $C_{\ell c}$ is the cycle matrix.

Adding a temporal component $t \rightarrow$ e.g. multiple points in time

$$\min_{g_{i,s,t}, f_{\ell,t}} \sum_s o_{i,s} g_{i,s,t}$$

such that

$$g_{i,s,t} \leq \hat{g}_{i,s,t} G_{i,s}$$

$$g_{i,s,t} \geq 0$$

$$\sum_s g_{i,s,t} - \sum_\ell K_{i\ell} f_{\ell,t} = d_{i,t} \qquad \text{Kirchhoff Current Law}$$

$$|f_{\ell,t}| \leq F_\ell \qquad \text{line limits}$$

$$\sum_\ell C_{\ell c} x_\ell f_{\ell,t} = 0 \qquad \text{Kirchhoff Voltage Law}$$

where $\hat{g}_{i,s,t}$ can, for instance, represent the capacity factor of a wind turbine.

Adding a temporal component $t \rightarrow$ **now with storage**

Add constraints where storage units r can dispatch power within its **discharging capacity**:

$$0 \leq g_{i,r,t,\text{discharge}} \leq G_{i,r,\text{discharge}}$$

and consume power to charge the storage within its **charging capacity**:

$$0 \leq g_{i,r,t,\text{charge}} \leq G_{i,r,\text{charge}}$$

Also add a constraint on the limit $E_{i,r}$ on the **state of charge** $e_{i,r,t}$:

$$0 \leq e_{i,r,t} \leq E_{i,r}$$

The dispatch decisions are related by the following constraint:

$$e_{i,r,t} = \eta^{\text{self-discharge}}_{i,r,t} e_{i,r,t-1} + \eta^{\text{charge}}_{i,r} g_{i,r,t,\text{charge}} - \frac{1}{\eta^{\text{discharge}}_{i,r}} g_{i,r,t,\text{discharge}}$$

One more thing: Shadow prices in optimisation problems

Shadow prices (aka dual variables or KKT multipliers) measure the marginal change in the objective value of the optimal solution by relaxing the constraint by a small amount.

Let's return to the **optimisation problem** for the merit order now including dual variables:

$$\min_{g_s} \sum_s o_s g_s \qquad s.t. \tag{5}$$

$$g_s \leq G_s \qquad \leftrightarrow \bar{\mu}_s \tag{6}$$

$$-g_s \leq 0 \qquad \leftrightarrow \underline{\mu}_s \tag{7}$$

$$\sum_s g_s = d \qquad \leftrightarrow \lambda \tag{8}$$

The **dual variable** λ corresponds to the **market clearing price**, i.e.:

- the marginal cost of the marginal generator
- the cost of consuming an additional unit of power